

ISRAEL AGUILAR ORDONEZ

COMPUTATIONAL BIOLOGIST & DATA SCIENTIST

Excellent background in **Human Genomics and Microbiology**

Expert in high-throughput data analysis for the genomic era



ORCID

ResearchGate



GitHub

Google Scholar

• CONTACT INFO



iaguilaror@gmail.com



Located at CDMX, Mexico

References



Dr. Victor Treviño
vtrevino@tec.mx

Dr. Enrique Morett
enrique.morett@ibt.unam.mx

• EDUCATION

PhD in Biochemical Sciences & Master in Sciences

Instituto de Biotecnología - UNAM

• BIOINFORMATICS SKILL SET

- Advanced **Python, R, SQL**, and **Bash**
- **CI/CD**, **git + github**, **REST**
- **ML algorithms**, **LLM**, **Data Mining**
- **AWS** and **Azure**
- **Nextflow** pipeline dev with **Docker**
- Specialist in **SingleCell RNA-Seq**, **Variant Calling**, **CNV**, **GWAS**, **DGE analysis**, & **Metagenomics**
- Fluent in **high-throughput computing** envs
- Hardware setup & **Linux SysAdmin**

• PROFESSIONAL EXPERIENCE

Posdoctoral Researcher

Tec. Monterrey, México

2 years (2024 - currently)

- Lead Computational Biologist for the oriGen project
- Statistical analysis of 100,000 exomes
- In charge of ancestry, kinship, and CNV-association studies

Head of Supercomputing Department

Instituto Nacional de Medicina Genómica, Mexico

5 years (2019 - 2024)

- In charge of the administration and maintenance of high-throughput computer systems
- Data analyst for key institutional projects

Interim Head of Bioinformatics

Instituto Nacional de Medicina Genómica

1 year (2020)

- Guide the design and deployment of new pipelines

Chief of Bioinformatics

Winter Genomics

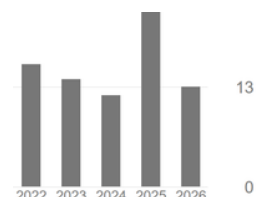
3 years (2015 - 2018)

• RESEARCH PROFILE

Cited by

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	All	Since 2021
Citations	99	89
h-index	5	5
i10-index	4	3



RESEARCH

1st or Corresponding

- **Aguilar-Ordoñez**, et al. "Comprehensive Whole Genome Sequencing of 1,427 Mexican Individuals: Insights from the oriGen Cohort". *Nat. Comms.*, in press.
- **Aguilar-Ordoñez**, et al. "[A Tale of Native American Whole-Genome Sequencing and Other Technologies](#)." *Diversity* 14.8 (2022): 647.
- **Aguilar-Ordoñez**, et al. "[Whole genome variation in 27 Mexican indigenous populations, demographic and biomedical insights](#)." *PloS one* 16.4 (2021): e0249773.

Co-author

- Fraser, et al. "Ancestry, admixture, and pathogens in contemporaneous Neolithic farmers and foragers on the Island of Gotland". *Nat. Comms. Biology*, in press.
- Kuri-Morales, Pablo, et al. "[oriGen cohort: a Mexican population-based epidemiological and genomic research platform](#)." *J Epidemiol Community Health* (2025).
- Aparicio-Bautista, Diana I., et al. "[Label-free quantitative proteomics in serum reveals candidate biomarkers associated with low bone mineral density in Mexican postmenopausal women](#)." *GeroScience* (2023): 1-19.
- Swain, Jitendriya, et al. "[F-actin nanostructures rearrangements and regulation are essential for SARS-CoV-2 particle production in host pulmonary cells](#)." *Iscience* 26.8 (2023): 107384.
- Eagles, Nicholas J., et al. "[SPEAQeasy: a Scalable Pipeline for Expression Analysis and Quantification for R/Bioconductor-powered RNA-seq analyses](#)." *BMC bioinformatics* 22.1 (2021): 1-18.

MEDIA COVERAGE

Science communication

- **Aguilar,I.**, et al. (2021). [Conociendo el Genoma Indígena Mexicano: el proyecto 100G-MX](#). *Biotechnología en Movimiento ... UNAM*, 26, 8-15.
- **Aguilar-Ordonez,I.** (2015). [Pescadores de ácido Ribonucleico](#). *Biotechnología en Movimiento ... UNAM*, 3, 14-15.

Radio, TV and Newspaper

- [Colabora UAP en catálogo de variación genómica de los grupos indígenas](#). (2021) **Milenio**
- El estudio de genomas mexicanos [De eso se Trata, con Ricardo Cartas](#). (2021) **TV BUAP**
- [Sobre el premio Nobel de Medicina 2022](#). De eso se Trata, con Ricardo Cartas. (2022) **TV BUAP**
- [La Bioinformática al Servicio de la Ciencia](#). Pulso Saludable. Starts at 32:00. (2024) **Proyecto Radio Mx**

ACADEMIC MEETINGS

As speaker: ~15

As teacher: ~20